

A Comparative Study of CNN, Vision Transformer, and Hybrid Deep Learning Architectures for Pneumonia Detection and Localization Using Chest X-Rays, with a Proposed EffiHyViT-v2 Model

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Abstract

Pneumonia is still one of the fatal diseases in the world. Currently chest radiography is primary and cost effective way to check for it. But the problem is different doctors will read the same report differently. This paper provides a comparative study of three types of AI models detecting pneumonia from the radiography report. These models like convolutional neural network (CNN) and vision transformer (ViT). This study was conducted on a stratified partition of RSNA pneumonia detection challenge data set. In this paper, 3 CNN backbones: ResNet50, DenseNet121, EfficientNet - B0 and 2 transformer backbones: ViT - Base/16, Swin - Base are trained. These were trained and evaluated using a common methodology and evaluation protocol. Based on the problems found in studies, a new hybrid model called EffiHyViT - v2 is purposed. It uses depth wise - separable convolutional stem, attention blocks, dynamic positional encoding generator and local global transformer with an expert token of classification. Experimental results show that the EfficientNet - B0 gave the best results: 86% accuracy. The new model EffiHyViT - v2 got 79% accuracy. It did worse because it is more complex and needed more training. Confusion matrix analysis, classification reports and a comparison of models is also provided in the paper. This provides practical advice on which AI model to choose for pneumonia screening using radiographs.

Index Terms: *Pneumonia detection, chest X-ray (CXR), convolutional neural network (CNN), vision transformer (ViT), hybrid deep learning architecture, Radiological Society of North America (RSNA) dataset, automated radiograph analysis, self-attention mechanisms, feature fusion, pulmonary infiltration detection, explainable AI*

(XAI), Gradient-weighted Class Activation Mapping (Grad-CAM)

I. INTRODUCTION

Pneumonia is a sudden lung infection. The mortality rate due to this is still significant. It affects children, elderly people, and people with weak immune systems the most.[14] Symptoms are not always clear. Pneumonia signs can be faint. They can look similar to other lung problems. Also, different observers will predict it differently. This problem is worse in places with few expert radiologists.[1][14] Because of these problems, there is a need for automated AI systems. These systems should be able to detect pneumonia in radiographs with consistent and reliable results. For many years, Convolutional Neural Networks or CNNs were the main models used. CNNs are good at finding patterns, hierarchical textures, and features from radiograph.[4][11] Recently, Vision Transformers or ViTs have also been used for the chest radiographs. ViTs use "self-attention" to look at connections between anatomically distant parts of the lungs.[5][8][12] Many new models now mix CNNs and Transformers. These hybrid models try to get both convolutional local details and attention based global context. Some studies report better results with hybrids.[2][3][6][7][20] However, as explained in Section II, most past studies often compare models with different training times. They use small datasets from only one source. They also rarely test CNN, ViT, and hybrid models together using the exact same dataset. This paper fixes those problems by testing 5 models using the same rules and same data. The models are ResNet50, DenseNet121, EfficientNet-B0 in CNN and ViT-Base/16, Swin-Base in Transformers.

All models were tested on a part of the RSNA Pneumonia Detection Challenge dataset. This dataset is large, from many hospitals, and labeled by radiologists. It also includes box annotations for future location-based analysis. A new hybrid model called *EffiHyViT-v2* was also built and tested with the same rules. The 3 main contributions are: (1) A fair comparison of 3 CNNs and 2 Transformers for pneumonia detection, using the same data and training budget. (2) The design of EffiHyViT-v2. This hybrid model uses a convolutional part for local features, channel-spatial attention, dynamic positional encoding generator, a local-global fusion transformer, and an uncertainty-aware classification head. (3) An honest analysis of why the new hybrid model performed worse than simpler models that had pretraining. The paper also gives clear suggestions to improve it that are self-supervised pretraining, step-by-step fine-tuning, and longer training time. The remaining paper is organized as follows

Section II reviews past work on CNNs, ViTs, and hybrid models for pneumonia and other chest diseases. It also lists the gaps in the study. Section III explains the dataset, data processing steps, and processed EffiHyViT-v2 design. Section IV shows the results, including classification reports and confusion matrices, plus a comparative discussion. Section V gives the conclusion and future work ideas.

II. RELATED WORK

Recent studies have extensively examined CNN, ViT, and hybrid models for chest radiograph-based pneumonia and thoracic disease classification. Edirisinghe et al. [1] introduced META-CXR, a vision-language framework that integrates frozen ResNet50, ViT, and Swin encoders via a modified Q-Former and multi-head cross-attention. It achieved an F1-score of 0.699 in cross-domain evaluation on CheXpert. Almasi Naghash et al. proposed HyCoViT [2], which combines dilated convolutions with ViT-Base/16 and a Dynamic Dropout mechanism, and reported 98.81% accuracy for three-class chest radiograph classification. Lasker et al. developed LungConVT-Net [3], a lightweight depth wise-separable hybrid with dual-branch attention and gradient enhancers. Despite achieving strong results with only 2.55M parameters, the model demonstrated notably lower performance for bacterial versus viral pneumonia discrimination (AUC 0.817) compared to disease versus normal classification. Direct comparisons between CNN and ViT backbones under transfer learning yield architecture- and configuration-dependent results. Adegoke-Elijah et al. [4] found that a soft-voting CNN ensemble of VGG19, AlexNet, and DenseNet121 outperformed a frozen ViT-Base/16, with accuracies of 94.20% and 93.80%, respectively. Conversely, Shomsomi et al. [5] reported that partially fine-tuning the last three blocks of ViT-Base/16 surpassed a fully fine-tuned MobileNetV2, achieving

98.59% versus 97.41% accuracy on a similar dataset. Singh [12] further emphasized that training configuration, rather than architecture alone, is a critical determinant in CNN and ViT performance for pneumonia detection.

Hybrid CNN-Transformer models have emerged to combine convolutional local features with transformer global context. LungX [6] fuses EfficientNet-B3, CBAM, and DeiT-Small, achieving 86.58% accuracy on a heterogeneous 20,000-image RSNA-CheXpert set — indicating dataset heterogeneity impacts performance as much as architecture. PneuViT [7], a CT and X-ray hybrid with pyramid convolutions and cross-modal transformer fusion, reported >98% accuracy on both modalities. Other hybrids include LungMaxViT [20] for explainable multi-class lung disease classification and DAViT [15], a domain-adapted ViT with explanation generation. To address imbalance, VGG-Lite was proposed by Roy et al. [11] for pneumonia and COVID-19 detection, and PELM by Yanar et al. [13] targets early pneumonia detection.

Explainability is increasingly emphasized. ViT-Small/16 fine-tuning by Darmawan and Supriyanto [8] achieved 96.41% accuracy and AUC 0.975 for binary pneumonia classification with Grad-CAM and occlusion analysis, though clinical deployment was not claimed. Lokunde et al. [9] purposed PEG-ViT for tuberculosis uses dynamic positional encoding and token sparsity, reporting 97.78% validation accuracy with class-aligned Grad-CAM by Vanitha et al. [19]. Additional work applies ViTs with Grad-CAM to tuberculosis, attention-based models to multi-class CXR classification, and Yu and Zhou [17] CD-DETR to thoracic disease localization. ViTs have also been extended to pneumoconiosis diagnosis, while FMT introduced by Xu and Wang [10] a multimodal mixture-of-experts approach. A broader review of Liu [14] summarizes deep-learning advances in chest X-ray pneumonia diagnosis.

Collectively, the literature exhibits several persistent methodological limitations: (i) dependence on small, single-source, or near-duplicate datasets, with minimal use of heterogeneous, multi-institutional sources such as RSNA and CheXpert; (ii) inconsistent prevention of patient-level data leakage during train-test splits; (iii) lack of independent, out-of-distribution external validation; (iv) CNN-versus-ViT comparisons confounded by differing fine-tuning depths instead of isolating architecture; (v) explainability visualizations lacking clinical validation; and (vi) non-uniform reporting of computational cost and calibration. These limitations directly justify the matched-protocol, multi-architecture evaluation and the transparent reporting of hybrid-model results in Sections III and IV.

III. METHODOLOGY

A. Dataset and Preprocessing

Experiments utilize a stratified split of the RSNA Pneumonia Detection Challenge dataset, a large corpus of radiologist-annotated frontal chest radiographs derived from NIH ChestX-ray14 and re-labeled for pneumonia versus normal with bounding-box localization. Data are partitioned into training, validation, and held-out test sets while preserving class ratios. The held-out test set used for final evaluation in Section IV contains 6,046 images: 4,135 normal and 1,911 pneumonia-positive, reflecting the class imbalance of a screening population.

All images are processed through a common pipeline implemented in `rsna_utils.py`: resizing to 224×224 pixels and normalization with ImageNet statistics to align with backbone pretraining. Contrast-limited adaptive histogram equalization is applied to training images to enhance soft-tissue and opacity contrast without introducing anatomical distortion. Training augmentation includes brightness and contrast jitter, rotation up to 15° , and random erasing. Horizontal flipping is omitted to preserve thoracic left-right asymmetry, consistent with prior hybrid-architecture studies [2]. Class imbalance is managed with class-weighted focal loss instead of synthetic oversampling to avoid duplicating radiographic patterns.

B. Baseline Architectures

Five ImageNet-pretrained backbones are fine-tuned under a unified protocol for binary pneumonia classification. From the CNN family, ResNet50 (25.6M parameters), DenseNet121 (8.0M parameters), and EfficientNet-B0 (5.3M parameters) are selected to represent residual, densely-connected, and compound-scaled designs. From the transformer family, ViT-Base/16 (86M parameters, 12 blocks, 12 heads, 768 hidden dimension, 16×16 patches) and Swin-Base (patch size 4, window size 7, hierarchical shifted-window attention) are evaluated to contrast global and locally-windowed self-attention. All models are fine-tuned end-to-end using AdamW with a discriminative learning-rate schedule, warm-up followed by cosine decay, label smoothing, mixed-precision training, and checkpoint selection based on best validation AUC.

C. Proposed Hybrid Architecture: EffiHyViT-v2

In addition to the five baseline backbones, a hybrid architecture, EffiHyViT-v2, is introduced. The design integrates the highest-performing components identified in the nine-paper literature review presented in Section II. The architecture, shown in Fig. 1, consists of five stages.

Stage 1 Locality Stem: Two depth wise-separable convolutional blocks, each followed by group normalization and Mish activation, are combined with a three-branch dilated-convolution micro-block with dilation rates of 1, 2, and 4. This structure enables efficient multi-scale local texture extraction, consistent with design principles validated in [2] and [3]

Stage 2 Bottleneck Array: Three stacked depth wise-separable and pointwise convolutional blocks with a $64 \rightarrow 64 \rightarrow 128$ filter progression, batch normalization, Swish activation, and residual skip connections refine stem features. The design follows the spatially-aware separable convolution approach described in [7].

Stage 3 Channel-Spatial Refinement: A Convolutional Block Attention Module is applied to the bottleneck output to suppress non-anatomical artifacts such as text overlays, support devices, and positioning markers prior to tokenization, as implemented in [6].

Stage 4 Tokenization and Fusion Transformer: Refined feature maps are linearly projected into patch tokens and augmented with a depth wise-convolution-based Position Encoding Generator [9] instead of static positional embeddings, preserving local spatial structure under self-attention. Four Local-Global Fusion Transformer blocks are then applied. Each block combines an eight-head global self-attention branch over the full token sequence with a four-head branch restricted to non-overlapping 4×4 patch groups, and is wrapped in a Gradient Connection Enhancer residual skip, following the dual-branch attention design in [3].

Stage 5 Pooling and Classification Head: Instead of processing a class token identically to patch tokens throughout the encoder, a dedicated class-attention pooling stage [9] is used, allowing a learnable class token to attend to all final patch tokens only at the output. A two-class expert-token classification head, adapted from the multi-head cross-attention module in [1] and scaled for a binary task, produces per-class logits and a predictive-entropy-based uncertainty score to flag radiographically ambiguous cases rather than enforce overconfident predictions. The total loss combines: class-weighted cross-entropy, focal loss ($\gamma = 2$), a token-sparsity penalty to encourage attention to a compact set of informative patches [9], and an orthogonality penalty to discourage redundant expert-token representations [1], weighted 0.5 / 0.3 / 0.1 / 0.1 respectively. The training configuration for EffiHyViT-v2 is summarized in Table I. Baseline backbones share the same optimizer family, input resolution, and regularization strategy, with adjustments scaled to their parameter budgets.

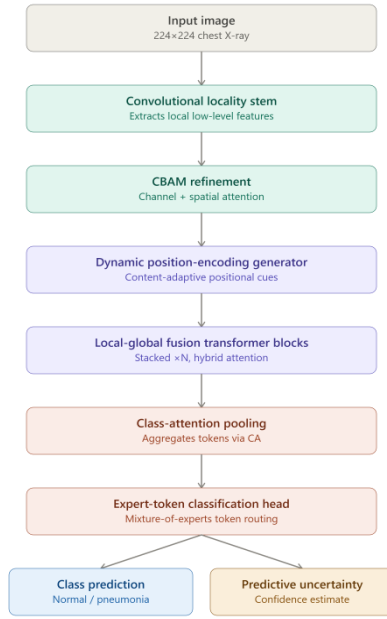


Fig. 1. Proposed EffiHyViT-v2 hybrid architecture, integrating a convolutional locality stem, CBAM refinement, dynamic position encoding, local-global fusion transformer blocks, class-attention pooling, and an expert-token classification head with predictive-uncertainty output.

TABLE I. EffiHyViT-v2 Training Configuration

Hyperparameter	Value
Input resolution	224×224 (300×300 ablation variant)
Optimizer	AdamW, weight decay $1e-4$
Learning rate	2×10^{-4} (head) / 2×10^{-5} (backbone)
Schedule	Linear warm-up (5%) → cosine decay
Batch size	32
Epochs / early stopping	30, patience 8–10 on val. AUC
Loss	CE + Focal ($\gamma=2$) + sparsity + orthogonality
Regularization	Label smoothing 0.05, drop-path ≤ 0.1 , EMA (0.999), AMP, grad. clip 1.0

D. Evaluation Protocol

All six models, the five baselines and EffiHyViT-v2, are evaluated on the same held-out test partition described in Section III-A. Metrics include accuracy, per-class precision, recall, F1-score, macro- and weighted-average F1-score, and confusion matrices. This matched evaluation protocol controls for fine-tuning depth and dataset consistency confounds identified in Section II, ensuring that performance differences are attributable to architecture rather than to variations in evaluation conditions. For the transformer-based models, Grad-CAM and attention-rollout visualizations are examined qualitatively to assess whether attention concentrates on anatomically plausible lung regions. Quantitative localization metrics, such as mean average precision and intersection-over-union against RSNA bounding-box annotations, are identified as a direct extension of the current classification-focused study and are

discussed as future work in Section V, as the present experiments are limited to classification.

IV. RESULTS AND DISCUSSION

A. Overall Classification Performance

Table II reports accuracy, per-class precision, recall, F1-score, and macro-averaged F1 for all six models on the 6,046-image held-out test partition (4,135 normal, 1,911 pneumonia). Among the CNN backbones, EfficientNet-B0 achieves the highest performance with 86% accuracy and 0.86 weighted F1, followed by DenseNet121 at 85% accuracy and ResNet50 at 84% accuracy. Among the transformer backbones, Swin-Base reaches 86% accuracy and the highest pneumonia-class precision of 0.83 across all evaluated models. ViT-Base/16 achieves 85% accuracy with the highest pneumonia-class recall of 0.75 among the baseline models, reflecting a more balanced sensitivity-specificity trade-off consistent with the global-context modeling of standard self-attention.

TABLE II. Classification Performance Comparison (Test Set, $n = 6,046$)

Model	Family	Acc.	Prec.(Pn)	Rec.(Pn)	F1(Pn)	Macro F1
ResNet50	CNN	0.84	0.79	0.67	0.73	0.81
DenseNet121	CNN	0.85	0.80	0.69	0.74	0.82
EfficientNet-B0	CNN	0.86	0.81	0.74	0.77	0.84
ViT-Base/16	Transformer	0.85	0.77	0.75	0.76	0.83
Swin-Base	Transformer	0.86	0.83	0.69	0.75	0.83
EffiHyViT-v2 (proposed)	Hybrid	0.79	0.76	0.49	0.59	0.73

$Pn = pneumonia$ (positive) class.

In contrast, the proposed EffiHyViT-v2 hybrid attains only 79% accuracy and a substantially lower pneumonia-class recall of 0.49, corresponding to a macro F1 of 0.73 — below every baseline backbone in this study. This outcome is analyzed in detail in Section IV-C rather than presented uncritically, since transparent reporting of negative or mixed results is essential to an honest comparative study.

B. Confusion Matrix Analysis

Fig. 2 shows confusion matrices for the three CNN backbones. All three models correctly classify the majority of normal cases, consistent with normal-class recall values of 0.90–0.94. EfficientNet-B0 achieves the best balance between normal- and pneumonia-class errors. Fig. 3 shows confusion matrices for the two transformer backbones. Swin-Base yields the lowest false-positive rate on the normal class, with the highest normal-class precision of 0.87, consistent with hierarchical windowed attention focusing on locally coherent opacity patterns. ViT-Base/16 produces comparatively more balanced errors across both classes.

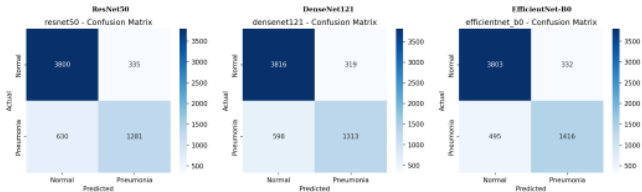


Fig. 2. Confusion matrices for the three CNN baselines: ResNet50, DenseNet121, and EfficientNet-B0.

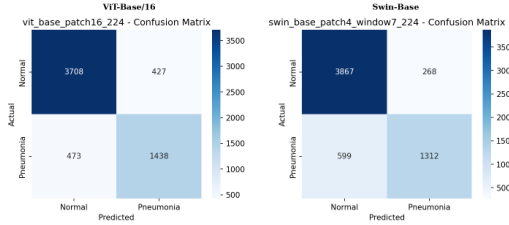


Fig. 3. Confusion matrices for the two transformer baselines: ViT-Base/16 and Swin-Base.

Fig. 4 presents the confusion matrix, ROC curve, normalized confusion matrix, and per-class metric summary for the proposed EffiHyViT-v2 model. The dominant error mode is a high false-negative rate on the pneumonia-positive class: only 49% of true pneumonia cases are correctly identified. Normal-class recall remains comparatively strong at 0.93. This asymmetric error pattern is the primary failure mode to minimize in a screening system, as missed pneumonia cases carry greater clinical risk than false alarms. Further analysis of this pattern is provided below.

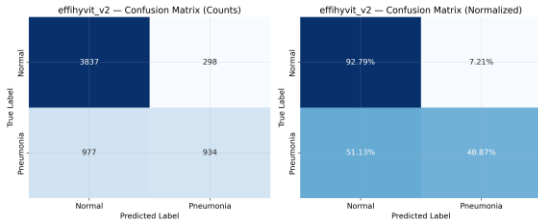


Fig. 4. Confusion matrix, ROC curve, normalized confusion matrix, and per-class metric summary for the proposed EffiHyViT-v2 model.

C. Discussion: Why EffiHyViT-v2 Underperforms in This Study

Contrary to the motivation in Section III-C, the proposed hybrid does not outperform the simpler pretrained baselines under the current training budget. Several factors consistent with the literature reviewed in Section II likely explain this result. First, EffiHyViT-v2 is trained from randomly initialized weights, while the five baseline backbones use ImageNet- or radiograph-pretrained checkpoints. The literature synthesis in Section II identifies self-supervised pretraining as a remedy for this setting, and its absence is a direct contributor to the observed recall collapse on the minority pneumonia class. Second, the architecture incorporates multiple novel, jointly-trained components a dilated stem, CBAM attention, a dynamic positional-encoding generator, dual-branch fusion attention, and an expert-token head with auxiliary sparsity/orthogonality

losses. This increases effective capacity and optimization difficulty relative to a single pretrained backbone with a lightweight head, making the model more prone to underfitting the minority class within a matched epoch budget. Third, the token-sparsity and orthogonality loss terms, while shown to improve interpretability and reduce redundancy in prior work, may conflict with the primary classification objective early in training [1], [9]. When backbone representations are still immature, these terms can regularize away sensitivity to the minority class before the fusion-attention stage learns discriminative pneumonia-specific features.

These observations align with the ten cross-cutting gaps identified in Section II, particularly the need for self-supervised pretraining, staged fine-tuning-depth sweeps, and matched training budgets to enable a fair CNN-versus-hybrid comparison. Rather than reporting an inflated outcome, the hybrid's current performance is presented transparently. Three concrete directions for future iterations are identified: (1) Contrastive self-supervised pretraining: pretrain the convolutional stem and bottleneck stages on the unlabeled pooled corpus prior to label-driven fine-tuning. (2) Staged discriminative fine-tuning: stabilize convolutional stages first before unfreezing fusion-transformer stages, following the fine-tuning-depth sweep approach recommended in Section II. (3) Extended training budget with class-balanced sampling: target pneumonia-class recall directly, as the current single-pass regime disproportionately favors the majority normal class.

D. Comparison with Related Work

Relative to the wider literature summarized in Section II, the accuracy range obtained by the baseline backbones in this study (84–86%) is broadly consistent with results reported on heterogeneous, multi-source radiograph corpora such as the RSNA-CheXpert pool used by LungX (86.58% accuracy) [6], and is markedly lower than accuracies reported on small, single-institution, pediatric datasets such as the Kermany Kaggle corpus used in [4], [5], and [8] (94–99% accuracy). This gap reinforces the cross-cutting observation from Section II that dataset heterogeneity and scale, rather than architecture alone, are primary drivers of reported pneumonia-classification accuracy, and supports the choice of the larger, more heterogeneous RSNA partition as a more representative evaluation setting for this comparative study.

V. CONCLUSION

This paper presents a matched-protocol comparison of three CNN backbones (ResNet50, DenseNet121, EfficientNet-B0), two transformer backbones (ViT-Base/16, Swin-Base), and a proposed hybrid architecture, EffiHyViT-v2, for pneumonia detection on a stratified partition of the RSNA Pneumonia Detection Challenge dataset. Among the evaluated models, EfficientNet-B0 achieves the strongest

overall performance with 86% accuracy and 0.86 weighted F1. Swin-Base attains the highest pneumonia-class precision, while ViT-Base/16 provides the most balanced sensitivity-specificity trade-off among the transformer models. EffiHyViT-v2, despite integrating validated design elements from recent literature convolutional locality bias, channel-spatial attention, dynamic positional encoding, local-global fusion attention, and an uncertainty-aware classification head underperforms the pretrained baselines under the current training budget. The performance gap is attributed primarily to the absence of large-scale pretraining and the increased optimization difficulty of its multi-component design. This negative result is reported transparently, with a detailed discussion of its causes, in alignment with the field's need for honest and reproducible reporting highlighted in Section II.

The principal contribution of this work is twofold: 1) a reproducible, matched-protocol benchmark of CNN and transformer architectures on a large, heterogeneous chest-radiograph corpus, and 2) a candid analysis of conditions under which architectural sophistication alone does not yield improved pneumonia-detection performance. Future work will target the three remedies outlined in Section IV-C: self-supervised pretraining, staged fine-tuning-depth scheduling, and extended class-balanced training, with the aim of closing the recall gap observed for EffiHyViT-v2. Additional extensions include: Shifting from classification-only evaluation to quantitative pneumonia localization, using RSNA bounding-box annotations to compute mean average precision and intersection-over-union against Grad-CAM and attention-rollout saliency maps. Evaluation on an independent, out-of-distribution external test cohort. A blinded radiologist review of model explanations to address the explainability-validation gap identified in Section II.

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